Agenda – 19 – Jan – 2025

- Model training in Yolo v5 (object detection)
- Yolo v2
- Evaluation Metric (Confusion matrix, Precision, Recall, Accuracy)

Confusion Matrix →

		Actual values	
		Positive	Negative
predicted values	Positive	TP	FP
	Negative	FN	TN

Dog, Dog → TP

cat, cat → TP

TP = GT is present, model predicted

TN → NO object, Model predicted None

FN → GT is present, model predicted None.

FP → Actual in None, model predicted

i/p → model → o/p

1. Dog, cat
2. Car, Car → TP
3. Mobile, Remote
4. Remote, Remote → TP
5. ___, Car → FP
6. ___, ___ → TN
7. car, ___ → FN

x_1, o, o, o, x_n → model → 23°C

→ 46

→ 16

→ 2

→ -1

MSE → $\frac{1}{2}(y'-y)^2$ →

TN → person, car, Dog →  → model → ☐

No object

None

Healthcare → Tumor

→ NO, Tumor

1|10

Tumor → NO →

FN

FP

23

100

A B

Doctor → 100 → 1/2

98 → No cancer

2 → cancer

slow, biased

98% 2%

48% 58% 100

$$\text{Accuracy} \rightarrow \frac{TP + TN}{TP + TN + FP + FN}$$

→ precision → $\frac{TP}{TP + FP}$ →

model predicted , Actual True

→ Recall → $\frac{TP}{TP + FN}$ →

Actual True , predicted car

→ F1 score → $\frac{2}{\frac{1}{recall} + \frac{1}{precision}}$

→ IOU

Hard hat
- images
  - test — images.jpg
  - train — im
  - val
- labels
  - test — image.txt
  - train
  - val

→ zip

Hard hat →

images —
- train
- val
- test

labels
- train
- val
- test

↔

Hard hat →

train
- images
- labels

val
- images
- labels

test
- images
- labels.

dataset

Yolo →

data →

yaml.
- path of data →
- train
- val
- test
- no classes & name

Step 1: we have cloned Yolo v5 repo

Step 2: Installation

Step 3: Downloaded dataset from roboflow

labels — a.txt

train < images — a.jpg

val

test

Step 4: Restructured dataset

Hardhat —
- images
  - train — a.jpg
  - val
  - test
- labels
  - train — a.txt
  - val
  - test

Step 5 Created data.yaml file

→ root directory of dataset

train: → images / train

val: → images / valid

test: → images / test

classes:

0 : person

1 : car

2 : Hat

Step 6 : Yolov5 / train.py

- yaml
- epoch
- weights
- device : 0

s → picked

m

l

x

pascal voc

COCO

CSV

txt

$x_1\ y_1\ x_2\ y_2$

$x_1\ y_1\ w_1\ h_1$

train →

train

val

test

v1

1000

val →

test

Team 1

Jan 2024 — V1

↳ left X

1000

v1

S3

→ 96%

→ 70%.

Team 2

↓

collecting

↓

data annotations → v2 1000 → v2 + v1 → train

yolo

model → val.py → 96% 98%.

not bad

70%.

10 epochs

epoch 1 → 0.03

epoch 3 → 0.01

epoch 6 → 0.02

epoch 8 → 0.005 — best.pt

epoch 10 → 0.03 → last.pt

train val →

2000

100

best →

1000

900

Datacollection

Jan 2025

→ google
→ pexels
→ Data scraping

$\boxed{2000}$ + 800

train val

test

train val

model.pt

test

Feb 2025

CCTV

Test

100
1000
800

100

→ train.py → model.pt → 3

→ detect.py → inference, plot bbox, video, images (yolov5.pt)
80

→ val.py → validation

→ export.py → pt → .engine
.onnx

resnet34
80 - 100

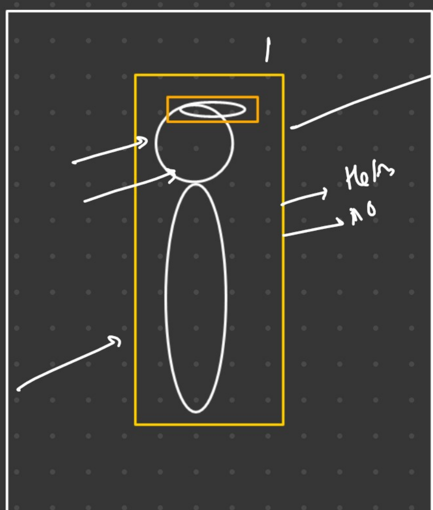
.jpg → [.jpg ; .png .]

Yolo → historical → Code

→ 2024 → ultralytics → ultra.plot. [Annotate]

→ yolo11

[yolov5 [v11] [v8]]



Yolo [person Helmet]

Helmet & person → FOV

[0.5%]

→ Helmets

[1 : No Helmet]

[yolo → person] → [Helmets]

→ ① → ② → Ref

→ CTO → 95